

Exam Code:2023031

BE 1st Year Examination
February March 2023
Electronics and Telecommunication (Section A & B)
COR2C5
Introduction to Computer Programming

Time: 3 Hrs

Max.Marks: 60

Note: Attempt any 2 parts of each question, all questions carry equal marks.

- 1
 - a) Compare and contrast between C language and C++ language.
 - b) Write a C++ program to print the sum of first n natural numbers. (n has to be entered by user).
 - c) Write a C++ program find out whether the number given by user is an Armstrong number or not.
- 2
 - a) Differentiate between Call by value and Call by reference with the help of swapping of numbers.
 - b) Write a C++ program to find out whether the entered number is prime or composite, using a function.
 - c) Write a recursive C++ program for printing Fibonacci series up to n terms (given by user).
- 3
 - a) Write a C++ program to check whether the entered string is palindrome or not using function.
 - b) Do as directed:
 - i. Declare structure "Distance" with data members: feet and inches. Declare pointer to the structure and use it for printing the distance as feet and inches.
 - ii. Declare an int pointer & float pointer. Dynamically allocate the memory using "new" operator. Assign the values, print them and then deallocate memory using "delete" operator.
 - c) Write a C++ program to find average of numbers in an array.
- 4
 - a) Describe in detail the concept of Friend function and Friend class with help of an example.
 - b) Write C++ code for creating class named "Shape", having overloaded function "area()" to calculate area of Shape Objects: rectangle, square, circle.

c) Write C++ code for creating class named "Account" for allowing to work like in Banking System and having:

- i. private data members: account_number, customer_name, account_balance, withdraw_amount, deposit_amount
- ii. public member functions: get_data (), deposit (), withdraw (), display ().
- iii. public parameterized constructor (for initializing account_balance=5000)

5 a) What do you mean by Inheritance? Explain with its types and example of each.

b) Write C++ code for creating class named "Complex" which allows addition of two complex numbers using the overloaded operator + (binary addition). Overloaded constructor should be used to assign/initialize the real and imaginary parts of complex number

c) Write C++ code for creating class "Student" publicly inherited from class "Person". The Class Person has:

- i. private data members: name, age, gender.
- ii. public member functions: getdata (), printdata ().

Class Student has

- i. private data members: grade, section, marks in 5 subjects
- ii. public member functions: getdata (), printdata (), result ().

Get name, age, gender, grade, section, marks from user, find the result and print it.
[Hint: Use of function over-loading.]